

Computational design and parent transcript liabilities of NR4A3 fusion junction gapmers in extraskeletal myxoid chondrosarcoma

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Running title. Parent pairing in NR4A3 junction gapmer designs

Abstract

Fusion-junction antisense gapmers could discriminate oncogenic transcripts from their normal parents, but partial parent pairing may escape conventional near-match screens. We computationally designed and screened 190 junction-spanning 16-mers across 38 in-frame NR4A3 fusion junctions in extraskeletal myxoid chondrosarcoma. Five sequence screens examined mature transcripts, exhaustive transcript substitutions, unspliced parents, contiguous mature-parent pairing through the DNA gap, and the genome. At an adopted ten-base-pair full-gap criterion, 87 of 190 designs (45.8%) paired a mature wild-type parent; the longest duplex involved NR4A3 for 61 designs. Exon-terminus chimeras met the same criterion at 40.6%, leaving a disease-specific excess unresolved. Within-junction selection found a parent-clear design for 35 of 38 junctions at this criterion, with substantial sensitivity to the chosen cutoff. Two designs at reported junctions had longest full-gap parent duplexes of eight and nine base pairs. The ten-base-pair criterion is a convention, not a measured cleavage threshold. Patient-derived cell-model junctions remain unresolved relative to these designs. The released

sequence records and screening outputs support reproducible computational prioritization and expose parent-transcript liabilities, but establish no cleavage, potency, delivery, safety or therapeutic window. No laboratory work was performed.

Keywords. antisense oligonucleotide; gapmer; RNase-H1; fusion transcript; NR4A3; extraskeletal myxoid chondrosarcoma; off-target screening

1 Introduction

Most EMC carries an in-frame *EWSR1::NR4A3* fusion[1]; *TAF15* is a substantial minority partner and *TCF12* and *TFG* are rare[2]. *FUS*, reported in two of five variant cases in a recent series[3], supplies eight modelled junctions here. Conventional cytotoxic chemotherapy has limited activity[4], though responses occur[5]. Disease control exceeded response with a trialled tyrosine-kinase inhibitor: this interpretation uses the review's response categories[4], not a figure stated in the trial report[6].

A fusion-junction gapmer could recruit RNase-H1 to cleave tumour RNA while sparing the normal parents. Junction-directed nucleic-acid agents have been reported against at least six fusion oncogenes: two as antisense oligonucleotides and the remainder as RNA-interference agents, including one expressed from a lentiviral vector[7-12]. None against an *NR4A3* fusion was found in the literature retrieved.

Each parent supplies roughly half the junction sequence. Such matches often fall outside a conventional off-target search's mismatch budget but may pair the catalytic gap. We therefore screen that pairing directly. RNase-H1's reported minimum DNA gap is six nucleotides, with seven to ten the working range[13]. Our criterion instead concerns a ten-base-pair contiguous duplex through the whole six-nucleotide gap. Ten is an

adopted hybridisation criterion, not a measured cleavage threshold or a gap-length requirement. We also adopt, without establishing, the premises that incomplete overall pairing can permit cleavage and that wing mismatches are less protective than gap mismatches. Of the five steps in a 2025 industry off-target framework[14], this work performs only the in-silico half of step one; transcriptomics is not performed. The Discussion proposes a parent-selectivity measurement related to step three.

2 Materials and methods

All analyses use public data; no laboratory work was performed. Complete parameters, per-design tables and claim bounds are archived. Canonical transcripts of five partners and *NR4A3* were obtained from Ensembl[15]. We tiled 16-mers in 5'-6-5 β -D-oxy-locked-nucleic-acid/DNA/ β -D-oxy-locked-nucleic-acid geometry[13], giving five registers per junction with the breakpoint inside the DNA gap. Six is the cited minimum, not the preferred gap length; the genome-wide arm is available only for 16-mers. Gap-level margin counts junction-unique gap bases on the shorter side of the breakpoint.

Five screens examine human RefSeq RNA alignments; an exhaustive transcript substitution search within one mismatch; unspliced parents; the longest contiguous duplex through the gap in six mature wild-type parents; and unambiguous GRCh38, including mitochondrial sequence. This adopted scope covers mature, precursor, exon-exon, non-coding and mitochondrial sequence. A near-match pairs at least 14 of 16 positions. Parent liability requires a contiguous run of at least ten base pairs pairing the whole gap. Ten identically screened null ensembles comprise four shuffles, two base-frequency draws (uniform or composition-matched), and four real-parent chimeras, two joined at exon termini.

Alignments were re-scored by nearest-neighbour stability of the longest contiguous paired run; only energy separations are reported. Melting-temperature calculations assume unmodified DNA:RNA at 250 nM strand[16], not locked phosphorothioate chemistry. Accordingly, we report no absolute melting temperature for the proposed reagents. Table 1 gives the unmodified-hybrid model's fusion-versus-parent difference. LNA and phosphorothioate effects are unmodelled; this difference is not a validated bound for the modified chemistry.

3 Results

3.1 Designs at reported junctions

We select the highest-margin designs at the two most-prevalent junctions with published exon-resolved breakpoints: 5'-GGGCATATCATCAAAC-3' for *EWSR1* exon 12 and 5'-GGGCATATCTTGTGTG-3' for *TAF15* exon 6, both to *NR4A3* exon 3 (Table 1). Exon indices count from the transcript 5' end, including non-coding exons; coding-exon numbering can select a different reagent. Both have gap-level margin three. Their longest wild-type duplexes pairing the whole gap are eight and nine base pairs, respectively, both against *TFG*. Thus both are liable at an eight-base-pair cut, the *TAF15* reagent remains liable at nine, and neither is liable at ten. Both also pair part of the gap at the wild-type *NR4A3* exon-2/exon-3 seam; these unmeasured partial duplexes are not counted. A one-base register shift can reverse the verdict: 5'-AGGGCATATCTTGTGT-3', adjacent to the *TAF15* reagent, pairs 11 base pairs of wild-type *NR4A3* through the whole gap and cannot substitute for it.

At a deeper-than-default search ceiling, the *EWSR1* and *TAF15* reagents have 123 and eight gap-paired sense-strand transcript near-matches, but six and five gene loci. Most of the 123 are predicted transcript models. The *EWSR1* reagent also matches wild-type *TAF15* precursor RNA across an intron-exon boundary with two mismatches, one in the gap. Ten shared donor bases let this reagent span *EWSR1*, *TAF15* and *FUS* breakpoints (Figure 1). The *TAF15* reagent has no sense-strand precursor site.

Aggregate genome-wide loads are similar: 156 and 135 hybridisable gap-paired sites. The *EWSR1* reagent has one exact 16-base genomic match, intronic in annotated lncRNA ENSG00000304430; the *TAF15* reagent has none. These are sequence predictions, not measured activity. Much genomic sequence is untranscribed; transcription and cleavage at these sites are unmeasured.

Both reagents are phosphorothioate throughout, with five contiguous β -D-oxy-locked residues per wing, exceeding the two to four taken here as usual. Their high affinity may permit more mismatches than the near-match screens' two-mismatch ceiling. Both begin with a locked 5'-GGG tract. Increased sequence-dependent hepatotoxicity with high affinity is an adopted, untested premise here, not a retrieved finding. Neither reagent's potency or safety is established.

Combining the breakpoint distribution of an 18-case series[17] with a 58-case cohort[18] assigns 68.4% of molecularly confirmed cases to these two junctions. This is modelled coverage, not patient screening. Its 39.9%–82.8% range is not a confidence interval: two of four inputs are fixed, no nominal level is assigned, and the calculation transfers a breakpoint distribution between cohorts collected 21 years apart. The *TAF15* arm uses three of three reported breakpoints, an upper bound. Adding the top-margin *EWSR1*

exon-13/*NR4A3* exon-3 design, 5'-GGGCATATCTCCACGG-3', would raise modelled coverage to 79.0%; selection names only the first two junctions by reported prevalence.

3.2 Selection and parent liabilities across 190 designs

Across 38 in-frame junctions of five partners, 87 of 190 designs let a mature wild-type parent pair the whole gap over at least ten contiguous base pairs. The longest duplex is with *NR4A3* for 61; 85 pair one of their own two parents. Only seven of the 87 reach the conventional 14-of-16 near-match threshold, and parent records are excluded by name from that screen. Nineteen designs have sense-strand precursor near-matches pairing the whole gap. The mature-parent and precursor union is 93 designs, not their sum. Gap mismatches score zero, so these counts bound the fully paired class, not all parent liability.

Alignment-screen cleanliness has additional bounds: seven searches never returned and censoring leaves only 47 of 183 returned searches assessable. A clean count is a floor over that subset; most default-clean designs fail a deeper search. Parent, precursor and genome screens cover all 190 without failures or censoring.

Energy re-scoring finds eight designs with fully paired 16-base off-target duplexes and 45 with a duplex within 2 kcal/mol of the intended target. Neither named reagent is in these classes; their nearest separations are 3.2 and 3.0 kcal/mol. These are upper bounds because the longest-run calculation ignores pairing on either side of a mismatch.

Longer gaps retain parent liability: counts are 87, 88 and 87 in panels of 190, 266 and 342, respectively, reducing rates from 45.8% to 33.1% and 25.4% without removing the problem. At 5-10-5, whole-gap pairing itself requires ten base pairs; shorter runs

potentially compatible with the reported seven-to-ten working range are missed, making 25.4% a floor.

Three designs clear every screen at the ten-base-pair cut. Two have no full-gap parent duplex at any length; the third pairs wild-type *NR4A3* over eight bases. None targets a reported patient junction, so these are mechanism controls. Within-junction selection instead finds a parent-clear design for 35 of 38 junctions and all five with published exon-resolved breakpoints. At cuts of nine, eight, seven and six base pairs, those counts become 31/3, 23/2, 9/0 and 6/0 (all junctions/published junctions). These cuts measure whole duplexes, not DNA gap length; the named *TAF15* junction loses its best design at nine.

Exon-terminus chimeras meet the parent-liability criterion at 40.6%, versus 45.8% for the panel. The strongest null falls inside the panel's 95% interval at cuts seven through thirteen except eleven; the excess changes sign four times across cuts six through thirteen. Both panel and chimeras contain mostly junctions unreported in patients. The comparison therefore does not resolve a disease-specific excess over the liability generated by joining these genes' exon termini.

3.3 Computationally screened control sequences

Table 2 gives a dinucleotide-preserving scramble of each reagent, screened against mature parents. Among such scrambles, 10.0% pair a parent's whole gap over at least ten bases and 3.9% have the longest duplex with *NR4A3*. Passing this screen is necessary for these controls; it does not establish inertness.

4 Discussion

4.1 Candidate models and unresolved junction correspondence

Five test articles offer complementary routes. The panel joins donors to *NR4A3* exon 3, its first coding exon, whereas two patient-derived cell models are reported at exon 2. No annotated *NR4A3* transcript has exon 2 as its first coding exon, so annotation does not reconcile them. The original protein-fusion filter excluded acceptors upstream of the initiation codon; that filter is inappropriate for RNase-H1 gapmers, which cleave RNA independently of reading frame. An *EWSR1* exon-7/*NR4A3* exon-2 fusion was sequenced in one of five *EWSR1*-rearranged tumours in a transcriptome series[19], and a *PGR::NR4A3* case joins exon 2 to the *NR4A3* 5' untranslated region[20]. All retrieved exon-resolved acceptors are exon 2, exon 3 or a cryptic exon in intron 2[21]; none is downstream of exon 3, where no patient-grounded designs are made.

Three engineered constructs come from a functional study reporting their exon spans[21]. Two, E-N and T-N*, carry exactly the named reagents' junctions. Rebuilding them offers a direct test of junction-selective knockdown, but heterologous complementary-DNA overexpression cannot establish activity at an endogenous locus.

The two patient-derived, identity-clean models are USZ20-EMC1 (RRID:CVCL_C6MX) and USZ22-EMC2 (RRID:CVCL_C6MY), the only fusion-positive EMC cell source identified here[22]. They are available on request, with no repository deposit and reported doubling times of five to six days. Their reported junctions join *EWSR1* exon 13 and *TAF15* exon 6 to *NR4A3* exon 2. The report supplies no sequenced boundary, transcript accession or junction sequence; whether this denotes

a non-coding acceptor or unsupported numbering cannot be determined. An earlier version of this work was withdrawn after this class of indexing error.

A first-coding-exon acceptor is more parsimonious for the defining chimeric transcription factor[1,21]. A non-coding exon-2 acceptor would instead retain the *NR4A3* initiation codon under the donor promoter. On the former interpretation, USZ22-EMC2 matches the named *TAF15* reagent and USZ20-EMC1 matches the third, exon-13 design. This remains inference. The builder emits exon-2 acceptors only for published patient seams or user-supplied sequencing checked against its transcript models.

Alternative exon-2 reagents are 5'-AGTGGGCTCTCCACGG-3' (*EWSR1* exon 13) and 5'-AGTGGGCTCTTGTGTG-3' (*TAF15* exon 6), both at top margin and below the ten-base-pair criterion. Their longest full-gap parent duplexes are eight bases with *EWSR1* and nine with *NR4A3*, respectively. The latter directly involves the acceptor used in the selectivity ratio. Reagents cannot be exchanged between acceptors.

Establish the test article's breakpoint at nucleotide resolution by RNA sequencing before ordering: only nine of 176 distinct panel sequences match multiple junctions. Routine break-apart *NR4A3* fluorescence in situ hybridisation detects rearrangement regardless of partner[4], not the nucleotide seam.

4.2 Proposed experimental evaluation

We propose isogenic fusion-positive/fusion-negative comparisons, as used for *NAB2::STAT6* in solitary fibrous tumour[23], with the screened controls. Knockdown alone cannot distinguish the relevant failure modes.

Selectivity is wild-type *NR4A3* knockdown IC50 divided by fusion knockdown IC50, from matched dose responses in the same wells. This ratio and its threshold of 5.0 are adopted conventions, not specified by framework step three[14]. With an assumed replicate standard deviation of 0.35 for the natural-log selectivity ratio, six independent biological replicates give about 80% power to falsify true selectivity of 3; three give about 30%. Variance is unmeasured.

Above a realised standard deviation of approximately 0.65 at three replicates, 1.53 at six or 2.25 at ten, no observed ratio at least one can place a two-sided 95% interval's upper limit below 5. Such a test can fail only with an anti-selective result and is treated as void. A normal-approximation interval would change these figures. Apply the gate to the upper confidence bound on a pilot's standard deviation: at or above the threshold for the proposed replicate count, increase replication or do not test. An acceptor-only ratio cannot exclude other parent liabilities; measure *TFG* alongside *NR4A3*, given its eight- and nine-base full-gap duplexes with the named reagents.

4.3 Interpretation and limits

Every modelled in-frame junction is designable; discrimination remains unproven. A design's own parents pair at most 13 bases across mature and precursor compartments, whereas eight off-target duplexes pair all 16, five in curated records. This is no liability ranking: the parent screen reads six transcripts and the energy screen excludes parents. Mature-parent liability requires a ten-base full-gap duplex; precursor liability requires a full-gap near-match within two mismatches. These are distinct, overlapping classes.

The archive additionally screens the patient's unrearranged *NR4A3* allele. Its two-mismatch ceiling gives a lower bound. No selected design fails, but two other registers at the *EWSR1* exon-13/*NR4A3* exon-2 seam do, reinforcing the register hazard.

Four cited junction-specificity reports tested synthesised molecules[9-11,24]; we did not survey design pipelines and claim no priority for pre-synthesis screening. Whether sparing wild-type *NR4A3* justifies a specificity cost remains unsettled: reported paralogue redundancy and dosage effects point in opposite directions.

All five screens predict hybridisation, not duplex formation or cleavage. Junction-directed oligonucleotides are established; the new application here is the indication. No potency, therapeutic window or safety has been measured, and systemic delivery to solid tumours remains unresolved. All named test articles require cell culture; none establishes clinical readiness.

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No person other than the author contributed to this work.

CRedit authorship contribution statement

Tristan D. McRae: Conceptualization, Project administration. The author conceived and directed the project and is responsible for its content. AI assistance with analysis and manuscript preparation is disclosed below.

Statements and Declarations

Research use only, and not for administration to any person or animal. Every sequence is an unsynthesised, untested laboratory reagent. Order only from fusion-

junction-aso-sequences.csv, which specifies sequence, locked residues and backbone, after RNA sequencing establishes the test article's breakpoint at nucleotide resolution.

Ethical considerations. Not applicable; no human subjects, human material or animals were involved, and no ethics approval was required.

Consent to participate. Not applicable; no participants were enrolled.

Consent for publication. Not applicable; no individual-level data are reported.

Declaration of conflicting interest. No competing financial interests exist. The author holds no patent, patent application, equity or consultancy relating to any sequence or method described here. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

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Data availability. Code, per-design tables and screen parameters for the preceding analysis are archived at [doi:10.5281/zenodo.22229096](https://doi.org/10.5281/zenodo.22229096). This revision does not claim that the archived files contain its corrected interpretations. The accompanying sequence record (Supplementary File 1) and revision note (Supplementary File 2) correct the melting-temperature interpretation and qualify cell-model correspondence; sequence rows and numerical model outputs are unchanged. An earlier analysis mislocated the acceptor through coding-versus-transcript exon indexing and was withdrawn in full. Rebuilt, verified panels and the correction record are archived.

Declaration of generative AI and AI assisted technologies

Use of artificial intelligence. Claude (Anthropic) assisted with analysis code, screens, literature retrieval and checking, and manuscript drafting and review. GPT-6-Astra (OpenAI) assisted with this revision and checks against repository artefacts.

Bibliographic records were retrieved from PubMed, Europe PMC or Crossref rather than generated, and citations were checked against those records. The author directed the work and is responsible for its content.

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Tables

Every value in the tables below is read from a committed artifact and none is typed by hand: the reagents from `fusion-junction-aso-sequences.csv`, the canonical machine-readable record, and the controls from `aso-control-oligos.json`, either directly or composed from their fields. The captions carry literature facts, each with its source named there. Every reagent named here is a 5-6-5 phosphorothioate gapmer. An oligonucleotide should be ordered from that file rather than transcribed from this page.

Table 1. The two reagents named for synthesis, with their parent-duplex label. The parent-duplex column is the longest contiguous duplex a mature wild-type parent forms through the catalytic gap; neither reagent reaches the ten-base-pair criterion, so the length is printed rather than a pass mark. The test articles are the engineered constructs of Brenca et al. ([PMID: 31020999](#)) — E-N for the *EWSR1* reagent and T-N* for the *TAF15* reagent; the two patient-derived models of Bangerter et al. ([PMID: 36316541](#)) are reported at an NR4A3 exon-2 acceptor; correspondence to these reagents requires nucleotide-junction confirmation. ΔT_m separates the fusion duplex from the more stable half of the design's own target window, which is a different parent from the duplex column's searched wild-type TFG. These are differences from the unmodified DNA:RNA model at 250 nM strand concentration. LNA and phosphorothioate effects are unmodelled, so these values are not validated predictions or bounds for the proposed modified reagents. Nothing here has been synthesised or tested, and no sequence may be administered to any person or animal.

Table 1

seam	reagent	margin	WT gap duplex (bp)	Model ΔT_m (°C)
<i>EWSR1</i> e12:: <i>NR4A3</i> e3	5'-GGGCATATCATCAAAC-3'	3	8 bp, wild-type <i>TFG</i>	26.6
<i>TAF15</i> e6:: <i>NR4A3</i> e3	5'-GGGCATATCTTGTGTG-3'	3	9 bp, wild-type <i>TFG</i>	36.0

Table 2. The two control oligonucleotides, each screened as its reagent was. Each is a dinucleotide-preserving scramble of the reagent it controls, matching it in length, first and last base, base composition and dinucleotide counts while spanning no junction, and each cleared the same mature-parent screen the reagent did. The Controls section above explains why that screening step makes a scramble a control.

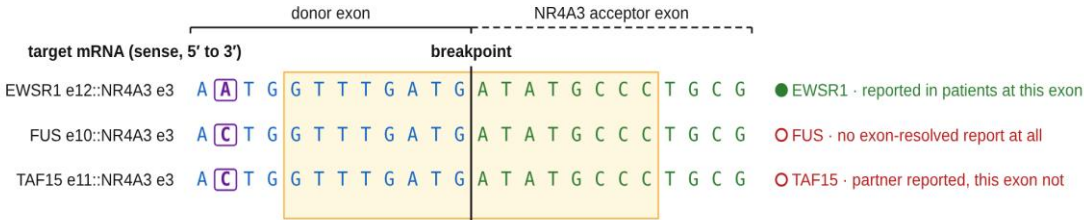
Table 2

control	sequence	scramble of	WT gap duplex (bp)
control-1	5'-GGGCATCAACATAATC-3'	the <i>EWSR1</i> e12 :: <i>NR4A3</i> e3 reagent	6 bp, wild-type <i>TAF15</i>
control-2	5'-GTATGTCATTGGCTGG-3'	the <i>TAF15</i> e6 :: <i>NR4A3</i> e3 reagent	7 bp, wild-type <i>EWSR1</i>

Figure legends

One 16-mer spans three partners' breakpoints — and only one of the three is a junction any patient is reported to carry

The three FET-family donors are identical over the ten nucleotides before the breakpoint, and the acceptor exon is the same in all three. Every row is TARGET mRNA, sense; the reagent is the reverse complement of the shaded window.



Solid rule and its label, the donor-exon columns; dashed rule and its label, the NR4A3 acceptor-exon columns; the two meet at the breakpoint. Boxed, the one position at which the three donors differ; filled marker, a seam reported in patients at this exon, open marker, one that is not. Shaded box, the window this reagent targets. Every one of those cues is drawn, so the panel reads in greyscale; online, the same donor bases are blue, acceptor bases green and the boxed position purple.

The reagent is 5'-GGGCATATCATCAAAC-3', the reverse complement of the shaded window (5'-GTTTGATGATATGCCC-3'). Transcribing the shaded letters orders the sense strand, which is a different molecule with no antisense activity.

Research use only, not for administration. The bases alone, ordered as unmodified DNA, are a different molecule; order from fusion-junction-aso-sequences.csv.

Figure 1. One 16-mer spans three partners' breakpoints; only one is reported in patients. Breakpoint-aligned windows join *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 to *NR4A3* exon 3, with reporting status for each row. The *TAF15* exon-11 row differs from Table 1's exon-6 reagent: it is an additional breakpoint spanned by the *EWSR1* reagent, not a selected reagent.